

STACK

PROGRAM CODING:

```
#include<stdio.h>

#include<conio.h>

#include<stdlib.h>

void main()

{

    int choice,n,a[30],top=0,i;

    clrscr();

    printf("\n Stack using array");

    do

    {

        printf("\n Main menu");

        printf("\n1.stack max");

        printf("\n2.push");

        printf("\n3.pop");

        printf("\n4.display");

        printf("\n5.size");

        printf("\n6.exit");

        printf("\n Enter your choice:");

        scanf("%d",&choice);

        switch(choice)

        {

            case 1:

                printf("\n Enter the stack max size:");
```

```
scanf("%d",&n);

break;

case 2:

if(top>=n)

{

    printf(" The stack is full");

}

else

{

    printf("\n Enter the element:");

    top++;

    scanf("%d",&a[top]);

}

break;

case 3:

if(top==0)

{

    printf("\n The stack is empty");

}

else

{

    printf("\n Poped out element is %d \n",a[top]);

    top --;

}

break;

case 4:
```

```
if(top!=0)
{
    printf("\n Stack contains \n");
    for(i=1;i<=top;i++)
        printf("\t %d",a[i]);
}
else
{
    printf("\n Stack is empty");
}
break;
case 5:
printf("\n The size of stack is %d",top);
break;
default:
    exit(1);
}
}while(choice<6);
getch();
}
```